

The “Astonishing Regularity” Revisited: Sensitivity of Learning-Rate Estimates to Practice-Sequence Length

Hansol Lee
Stanford University
Stanford, CA, USA
hansol@stanford.edu

Guilherme Lichand
Stanford University
Stanford, CA, USA
glicheand@stanford.edu

Cristina Barnard
Stanford University
Stanford, CA, USA
crisbg@stanford.edu

Lucas Klotz
University of California San Diego
La Jolla, CA, USA
lucasiklotz@gmail.com

Candace Thille
Stanford University
Stanford, CA, USA
cthille@stanford.edu

Yunsung Kim
Stanford University
Stanford, CA, USA
yunsung@stanford.edu

Benjamin W. Domingue
Stanford University
Stanford, CA, USA
bdomingu@stanford.edu

Abstract

A 2023 *PNAS* study by Koedinger et al. [20] fit the individual Additive Factors Model (iAFM) to 27 educational datasets and reported an “astonishing regularity” in student learning rates: students vary substantially in initial knowledge but learn at remarkably similar rates with practice. We probe a largely unexamined assumption underlying this finding—that observation length in student log data is ignorable for mixed-effects estimation—by refitting the iAFM on 26 of the original datasets while systematically truncating practice sequences at various depths, holding the set of students and knowledge components constant. Capping at the first ten opportunities per student–skill pair inflates the median estimated IQR of student learning rates by 75%; capping at five inflates it by 205%, with individual datasets ranging from negligible to 17-fold. The magnitude of this sensitivity diverges from what standard estimation theory predicts under ignorable truncation, and the dataset-specific heterogeneity is substantial. Three candidate mechanisms from adjacent literatures could account for the pattern—informative observation length, functional-form misspecification, and identification weakness from sparse per-pair data—but observational analysis on these data alone cannot adjudicate among them. We argue that practice sequence length distributions are an unexamined property of mixed-effects estimation on observational learning data, deserving explicit reporting before conclusions about learning-rate heterogeneity are drawn.

CCS Concepts

• **Applied computing** → **Education**; • **Social and professional topics** → **Student assessment**; • **Computing methodologies** → **Modeling and simulation**.



This work is licensed under a Creative Commons Attribution 4.0 International License. *L@S '26, Seoul, Republic of Korea*.

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2293-6/2026/06
<https://doi.org/10.1145/3774398.3811619>

Keywords

learning analytics; mixed-effects models; iAFM; learning rate; variance components; observation structure; practice-sequence length

ACM Reference Format:

Hansol Lee, Guilherme Lichand, Cristina Barnard, Lucas Klotz, Candace Thille, Yunsung Kim, and Benjamin W. Domingue. 2026. The “Astonishing Regularity” Revisited: Sensitivity of Learning-Rate Estimates to Practice-Sequence Length. In *Proceedings of the Thirteenth ACM Conference on Learning @ Scale (L@S '26)*, June 29–July 3, 2026, Seoul, Republic of Korea. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774398.3811619>

1 Introduction

How much do students vary in their rate of learning? A 2023 *PNAS* study by Koedinger et al. [20] offered a striking answer: while students differ substantially in initial knowledge, their rates of improvement with practice are remarkably similar. Analyzing 1.3 million practice interactions from nearly 7,000 learners across 27 diverse datasets using the individual Additive Factors Model (iAFM [7, 24]), the authors concluded that achievement gaps primarily reflect differences in prior preparation and access to learning opportunities rather than differences in learning efficiency. The implications are direct: if learning rates are genuinely uniform, educational disparities arise from where students start, not from how they progress. This interpretation stands in tension with longstanding theoretical accounts that posit “Matthew effects” or cumulative advantage in learning [16, 35], in which initial differences compound over time through differential rates of skill acquisition. Whether learning rates are genuinely uniform across students, or whether observed regularities reflect features of the data and estimation procedure rather than the underlying population, therefore matters for how we understand the mechanisms behind educational inequality.

Several robustness checks accompanied the original finding. The pattern replicates in independent datasets [31]; bootstrapped confidence intervals around variance components are tight; and simulation studies confirmed that the iAFM can recover learning-rate variability when it exists at substantially amplified levels [20]. Together,

these checks establish that the iAFM can detect heterogeneity when it is present.

A separate question, however, has not been examined empirically. The variance-component estimates that produce the regularity finding are computed by a mixed-effects model fit to observational student log data, where how much each student practices a given skill is not experimentally controlled—it emerges from the interaction of system design, student behavior, and instructional context. The model treats this observation structure as ignorable: it assumes the number of attempts a student gets on a skill is unrelated to how that student would have performed on additional attempts. When this assumption fails, the quantity the model is trying to estimate—how much students differ in learning rate—can become entangled with how much of each student’s trajectory the data happen to capture.

To make the issue concrete, consider a mastery-based intelligent tutoring system in which a student practices a skill until producing three correct responses in a row. A student who learns quickly demonstrates mastery in a few attempts; a student who learns slowly takes many. An analyst then fits an iAFM to log data from the system. The fast learner contributes a few right/wrong responses on this skill; the slow learner contributes many. The model gives the slow learner’s flatter trajectory more weight, so the estimate of how much students vary in learning rate depends in part on which students stayed in long enough to be observed. Similar dependences can arise through other channels. Students may extend sequences through wheel-spinning [5] or shorten them through disengagement [2, 3] rather than mastery. The iAFM assumes a linear relationship between practice opportunities and the log-odds of a correct response, an assumption that may hold over short stretches of practice but not longer ones [12, 14, 25], so that the slope estimated for any student depends on which portion of the underlying learning curve the data covers. And when each student contributes only a handful of practice attempts per skill—as is typical in these data—the model has limited information from which to estimate per-student learning rates separately from the population average [24]. Each of these channels ties what the model concludes about students to how much of each student’s trajectory was observed.

We examine the implications using the same 26 datasets and modeling framework as Koedinger et al. [20]. Practice sequence lengths in these data are highly imbalanced: some students contribute only a handful of attempts on a given skill while others contribute hundreds, and the shape of this imbalance varies substantially across datasets. Estimated student-slope variability is substantially sensitive to which observation depth is included in estimation: refitting with sequences capped at the first ten practice opportunities increases the median estimated IQR of student learning rates by 75%; capping at five increases it by 205%, with individual datasets ranging from negligible to 17-fold. The empirical magnitude diverges substantially from what standard estimation theory under ignorable truncation would predict.

Practice sequence length distributions shape what mixed-effects models conclude about learning rates on observational data, but rarely receive attention in the standard practice of fitting and reporting these models. Our contribution is to document this dependence empirically across the 26 datasets, and to connect what we find to

three candidate accounts from adjacent literatures—longitudinal data analysis, the functional form of learning curves, and identifiability of individual-level parameters—each of which could plausibly produce the empirical pattern we observe. The methodological point stands regardless of which mechanism is operative: conclusions about how much students vary in learning rate, drawn from estimates of this kind, depend on the observation structure of the underlying data.

2 Background and Related Work

We situate our work in two literatures: empirical work on student log data and learning analytics, which characterizes how practice sequences are generated in real systems and what is known about estimation in this setting; and statistical work on longitudinal data analysis, which provides conceptual frameworks for several of the challenges that arise when estimating models on data of this kind.

2.1 Variable Practice Depth in Observational Student Log Data

In observational student log data, practice depth—the number of opportunities a student accumulates on a given knowledge component—is not experimentally controlled. It emerges from the interaction of system design, student behavior, and instructional context.

System designs link exit to performance in several ways. Mastery-based learning systems [6, 13] advance students once they demonstrate competence, so students who learn more quickly naturally generate shorter practice sequences on a given skill. Adaptive sequencing in intelligent tutoring systems routes students to additional practice on skills where they perform poorly [19, 33], with similar effect. Open-ended practice environments shift the locus of control toward the student, so length variation reflects choices about when to stop, what to revisit, and how persistently to engage [1].

On the student side, behavioral phenomena shape practice depth even when system design is held constant. Wheel-spinning—the failure to improve despite extended practice—generates long sequences with flat learning trajectories [5]. Off-task behavior [2] and gaming the system [3] affect both the number and the validity of practice opportunities. Engagement and persistence covary with performance, difficulty, and motivational factors [2, 3], so length distributions in observational data carry traces of these behavioral processes.

These mechanisms together produce length distributions that vary substantially within and across datasets. In the dataset collection examined by Koedinger et al. [20], median student–KC pair lengths range from 2 to 21 opportunities, with within-dataset standard deviations from 1.15 to 53.85 (Section 4.1). Liu and Koedinger [24], in introducing the iAFM, observed that dataset depth is a critical factor in whether the model recovers reliable individualized estimates, and that sparse data degrade parameter recovery. The within-dataset *distribution* of depth across student–KC pairs is the structural feature our analyses focus on.

2.2 Estimation Challenges in Mixed-Effects Models on Observational Learning Data

Mixed-effects models applied to observational longitudinal data make implicit assumptions about both the outcome process and the observation process. Three estimation challenges relevant to the analysis below have been documented in adjacent literatures, and each provides a candidate account of how variance-component estimates could depend on the structure of the data.

Ignorability and informative observation length. Maximum-likelihood estimation in mixed-effects models requires that the missing data mechanism be ignorable: the probability of observing a given sequence length must be independent of unobserved outcomes (missing completely at random, MCAR) or depend only on observed covariates (missing at random, MAR) [23, 30]. When continued observation depends on the unobserved trajectory itself—roughly, when how long a student practices depends on how that student is actually performing—the missing-not-at-random (MNAR) or informative-observation-length condition holds [10, 22], and likelihood-based estimation can yield biased variance components. Under this condition, units with more observations exert greater influence over the likelihood and constitute a selected subpopulation whose trajectories differ systematically from the population’s [29]. Methods for principled estimation under MNAR include selection models [10, 37], pattern-mixture models [21], and joint longitudinal–time-to-event models [28, 32, 36]; comprehensive reviews are given in Hogan et al. [15] and Daniels and Hogan [9]. Despite this developed literature, informative observation length remains a commonly unexamined assumption in applied longitudinal research [23], and how it bears on student learning models has received limited attention.

Functional-form assumptions. Mixed-effects learning curve models also assume a specific functional form for the outcome trajectory. The iAFM, like the Additive Factors Model from which it derives [7], assumes log-odds of correct response are linear in practice opportunities. Whether learning curves have this form has been a long-running subject of debate: power-law and exponential alternatives have been proposed and contested across cognitive psychology and skill acquisition [12, 14, 25]. When the true curve is nonlinear and a linear specification is fit, the best-fitting linear slope for any segment depends on which portion of the curve the data covers—a misspecification that produces a different kind of dependence on observation length than informative observation length, but one whose empirical signatures can overlap.

Identifiability of individual-level parameters from sparse data. A third challenge is whether the data carry enough information to pin down individual-level random effects when each unit is observed only briefly. Käser et al. [17] showed that different parameter combinations can yield identical predictions in AFM-type learning curve models. Čechák and Pelánek [34] documented that item ordering biases systematically affect parameter estimates. Liu and Koedinger [24] explicitly observed that sparse data degrade individualized parameter recovery in the iAFM, and similar concerns have been raised for Bayesian knowledge tracing [11]. At the typical pair depths in this collection (median ≈ 6.5 opportunities), individual student-slope deviations b_{1i} may be poorly identified separately

from the population-mean slope β_1 and the KC-level structure. The resulting estimates are then dominated by the prior and the joint estimation rather than by individual student-slope information.

The implications of these challenges for population-level variance-component estimation in mixed-effects student models have received limited attention in the learning analytics literature; their joint implications, even less.

3 Methods

We examine how the variance-component estimates from the iAFM applied to the 26 datasets analyzed by Koedinger et al. [20] depend on the structure of practice sequence lengths in those data. After describing the data and model (Sections 3.1–3.2) and confirming that our implementation reproduces the original findings (Section 3.3), we describe our analysis strategy (Section 3.4): a characterization of the empirical setting, followed by a main analysis testing whether variance-component estimates depend on observation depth.

3.1 Data

Koedinger et al. [20] analyzed 27 datasets comprising over 1.3 million binary response observations from 6,946 learners across 12 courses, spanning multiple subject areas, educational levels, and instructional technologies, including intelligent tutoring systems, educational games, and online courses. The mechanisms generating practice sequence lengths differ across these system types, and the dependence we document may operate through different channels in different datasets.

We used the same knowledge component (KC) models and data reported in Table S3 of Koedinger et al. [20].¹ Of the 27 datasets, 26 were accessible; dataset ds4555 was restricted and could not be obtained. Observations with missing KC labels or responses coded as unknown were removed. This preprocessing yielded observation counts identical to those in the original study for 25 of the 26 datasets; the only discrepancy occurred in ds392, where 154 additional observations were excluded under the same criteria. Following the original study, we retained only KCs with data from at least 10 students and a minimum of two practice opportunities.

For each student–KC pair, we define its sequence length $L_{i,k}$ as the total number of practice opportunities student i had on knowledge component k .

3.2 The iAFM Model

3.2.1 Model Specification. The iAFM [24] models student performance as a function of prior practice opportunities on the knowledge components required by each task. Let Y_{ij} denote whether student i answered task j correctly on the first attempt, T_{ik} the number of prior opportunities student i has had on KC k , and q_{jk} the cognitive model mapping from tasks to KCs ($q_{jk} = 1$ if task j involves KC k , 0 otherwise). The model defines the log-odds of a correct response as:

$$\log \frac{P(Y_{ij} = 1)}{1 - P(Y_{ij} = 1)} = \beta_0 + b_{0i} + \sum_k q_{jk} c_{0k} + \sum_k q_{jk} (\beta_1 + b_{1i} + c_{1k}) T_{ik}, \quad (1)$$

¹Datasets were accessed through the PSLC DataShop [18] at <http://pslcdatashop.web.cmu.edu>.

where β_0 and β_1 are population-average fixed effects, b_{0i} and b_{1i} are student-level random intercepts and slopes, and c_{0k} and c_{1k} are KC-level random intercepts and slopes. The total slope for student i on KC k , which we refer to as the *pair slope*, is $\beta_1 + b_{1i} + c_{1k}$.

3.2.2 Estimation. Random effects follow zero-mean Gaussian distributions:

$$b_i \sim \mathcal{N}(0, \Sigma_{\text{student}}), \quad c_k \sim \mathcal{N}(0, \Sigma_{\text{KC}}). \quad (2)$$

Following the implementation of Koedinger et al. [20], Σ_{student} and Σ_{KC} are unstructured covariance matrices, allowing intercept–slope correlations to be estimated at both the student and KC levels. Fixed effects and variance components are estimated by maximum likelihood [27]; individual random effects are estimated as conditional modes (BLUPs [29]). BLUPs are shrinkage estimators: pairs with more observations are shrunk less toward the mean and contribute more to variance-component estimation. When sequence lengths vary substantially across pairs, this differential weighting becomes consequential—and the resulting variance-component estimates depend on whether the structure of that variation is well behaved.

Models were fit using `lme4::glmer` [4] with a binomial family and the `optimx` optimizer (method `nlminb`), following the procedures described in Koedinger et al. [20].

3.3 Replication

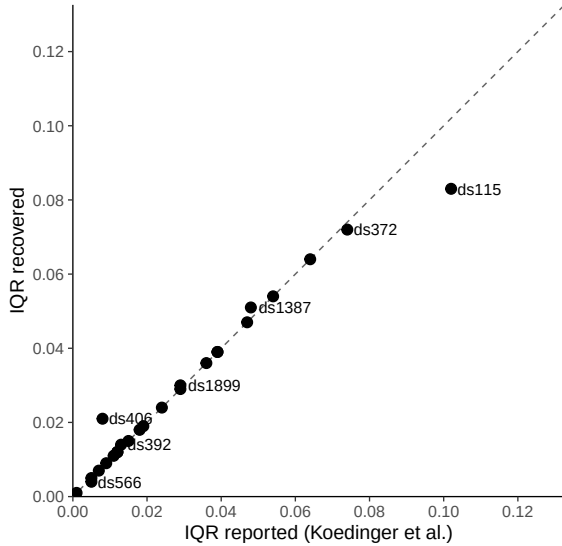


Figure 1: Replication of reported IQR of student learning-rate slopes from Koedinger et al. [20]. Each point represents one dataset; the diagonal indicates perfect agreement.

We first verified that our implementation reproduces the original findings. Figure 1 compares the IQR of student learning-rate slopes from our analysis to those reported in the original study; the close correspondence (with minor deviations attributable to small data

differences described in Section 3.1) confirms successful replication. Across 26 datasets, the median IQR of student learning-rate slopes is 0.020 ($M = 0.028$, $SD = 0.022$), compared to a median IQR of 0.136 ($M = 0.168$, $SD = 0.133$) for KC slopes and 0.797 ($M = 0.948$, $SD = 0.477$) for student intercepts. The central reported pattern is reproduced: large heterogeneity in initial knowledge and KC-level learning rates, but comparatively limited variability in student-specific learning rates.

3.4 Analysis Strategy

The main analysis tests whether variance-component estimates depend on observation depth. We first characterize the empirical setting.

We characterized the distribution of student–KC pair lengths within each dataset using the median (L_{med}) and standard deviation (L_{sd}), documenting the nature and degree of imbalance across datasets.

For the main analysis, we refit the iAFM on data restricted to sequences with $L \leq 10$ and $L \leq 5$ opportunities, comparing the resulting variance component estimates to those from the full model. This design holds sample composition constant: the same students, KCs, and student–KC pairs appear in all conditions, while varying only the depth of observation included in estimation. Because we retain the first k observations of each pair in their original order, the prior-opportunities counter T_{ik} takes its natural values $0, 1, \dots, k-1$ within each truncated pair, identical to its values for those same observations under the full-data fit. Under ignorable observation length, changing the observation window should not systematically alter what the model concludes about learning-rate heterogeneity. The $L \leq 10$ and $L \leq 5$ thresholds are substantively motivated: $L \leq 10$ roughly corresponds to the practice depth at which wheel-spinning has been characterized [5] and captures most pairs in most datasets; $L \leq 5$ shows whether sensitivity is more pronounced at shorter windows. To generalize beyond these binary comparisons, we also refit the iAFM across truncation windows $k = 5, 6, \dots, 20$.² As an exploratory follow-up, we examine how the sensitivity varies with dataset structure: we classify each dataset in the $L \leq 10$ comparison by whether truncation moves the estimated student-slope IQR above or below the full-data estimate, use Mann-Whitney U tests to assess whether structural features (L_{med} , L_{sd} , n_{pairs} , proportion of long pairs) distinguish the two groups, and compute the Spearman rank correlation between L_{sd} and the percent change in IQR. We report this follow-up descriptively, with appropriate caution given the small sample size ($N = 23$).

For all analyses, we report both standard deviations (SD) and interquartile ranges (IQR) of random effects to characterize heterogeneity, following the conventions of Koedinger et al. [20].

4 Results

Our main result is that variance-component estimates depend substantially on observation depth (Section 4.2). Section 4.1 first characterizes the empirical setting by documenting the practice-length distributions across the 26 datasets.

²Windows $k < 5$ were excluded due to potentially unreliable slope estimation at very short sequence lengths.

4.1 Practice-Length Distributions Are Highly Imbalanced

Figure 2 documents the practice-length distributions. Across the 26 datasets, L_{med} ranges from 2 to 21 (25th/50th/75th percentiles: 3 / 6.5 / 8) and L_{sd} ranges from 1.15 to 53.85 (25th/50th/75th percentiles: 3.3 / 9.0 / 18.9). Some datasets have low median lengths but high variance, indicating that while most pairs are short, a small number accumulate very long sequences (e.g., ds308: $L_{\text{med}} = 3$, $L_{\text{sd}} = 21.06$; ds566: $L_{\text{med}} = 8$, $L_{\text{sd}} = 29.54$). Others show both high median and high variance, indicating widespread extended practice (e.g., ds531: $L_{\text{med}} = 15$, $L_{\text{sd}} = 24.23$; ds1007: $L_{\text{med}} = 21$, $L_{\text{sd}} = 17.97$). Still others are tight and short throughout (e.g., ds115: $L_{\text{med}} = 2$, $L_{\text{sd}} = 1.15$). In all cases, the imbalance is a structural feature of the observational data rather than a design choice.

4.2 Variance-Component Estimates Depend on Observation Depth

Table 1: Median estimated learning-rate variability across datasets under different observation conditions. Values in parentheses are the percent change relative to the full-data baseline.

	Full data ($n = 26$)	$L \leq 10$ ($n = 23$)	$L \leq 5$ ($n = 25$)
SD	0.019	0.033 (+74%)	0.043 (+126%)
IQR	0.020	0.035 (+75%)	0.061 (+205%)

4.2.1 Truncation inflates estimated learning-rate heterogeneity. The same students, the same model, and the same knowledge components yield substantially different estimates of learning-rate heterogeneity depending on how much of each student’s practice trajectory is included. Table 1 reports the central result. Excluding sequences longer than ten opportunities increases the median IQR of student learning-rate slopes by 75%, from 0.020 to 0.035. Excluding sequences longer than five increases it by 205%, to 0.061. Individual datasets show increases ranging from negligible to 17-fold. The $L \leq 10$ comparison uses 23 datasets and the $L \leq 5$ comparison uses 25, with the others excluded because truncation at the relevant threshold leaves their data unchanged. Table 1 reports both the BLUP-based IQR (the regularization-sensitive empirical dispersion of estimated student slopes $\hat{\beta}_1 + \hat{b}_{1i}$) and the variance-component SD parameter ($\hat{\sigma}_{b_1}$) directly returned by the model fit; both move together under truncation, indicating that the sensitivity is at the level of the model’s variance-component estimates and not only their regularized BLUP summaries.

Figure 3 shows that the pattern holds across most datasets, not just in aggregate: 18 of 23 datasets fall above the diagonal for slopes, indicating higher estimated heterogeneity when extended sequences are excluded; five datasets show modest declines under truncation. Fixed-effect curves are stable across truncation while variance components move—a diagnostic that the effect is about heterogeneity, not central tendency or model collapse.

4.2.2 The magnitude diverges from what ignorable truncation would predict. A natural concern is that truncation reduces the number

of observations per pair, so some change in the spread of estimated random effects is expected regardless of mechanism. The question is whether the change we see is the kind one would expect from data reduction alone. Standard estimation theory provides a benchmark. If sequence length were independent of the random slopes, capping sequences at $L = 10$ would amount to MCAR-within-units for the affected pairs: their later observations would be dropped, but the dropping rule would not depend on their unobserved trajectories. Under that ignorable benchmark, the empirical IQR of BLUPs should be approximately stable or contract: the variance component remains approximately consistent while BLUP shrinkage *increases* as n_i shrinks (shrinkage scales roughly as $\sigma_e^2 / (\sigma_e^2 + n_i \sigma_b^2)$). The 75–205% inflation we observe diverges substantially from this prediction. We do not claim the empirical pattern alone rules out random data reduction as a contributor: a definitive empirical control would require constructing genuinely random observation subsamples while preserving the original opportunity indexing, which we do not do here. What we report is that the magnitude of the empirical sensitivity exceeds what standard theory under ignorable truncation would predict. We discuss candidate mechanisms—informative observation length, functional-form misspecification, and identification weakness at shallow depth—in Section 5.

Figure 4 reinforces this point at the level of the average trajectory: the fixed-effect learning curves from the full and truncated models are nearly identical, meaning truncation does not change what the model estimates about how the typical student improves with practice. The two models disagree only about how much students vary around that average, and that disagreement is large.

4.2.3 Sensitivity persists across truncation windows. The $L \leq 10$ and $L \leq 5$ thresholds in Table 1 are particular reference points; the broader question is whether variance-component estimates are stable across observation depths. To examine this, we refit the iAFM across truncation windows $k = 5, 6, \dots, 20$ for each dataset. At each value of k , only the first k observations from each student–KC pair are used; pairs with natural sequences shorter than k contribute their full data unchanged. Under ignorable observation length, the estimated IQR should be approximately stable across k . Figure 5 shows that it is not. As examples, ds394 (median pair length 5, $L_{\text{sd}} = 3.84$) shows the typical inflated pattern at short windows that gradually approaches the full-data baseline, while ds567 (median pair length 3, $L_{\text{sd}} = 2.81$) is one of the five datasets whose IQR contracts under short windows rather than inflating. The shape of the IQR-versus- k curve also differs substantially across datasets, which prompts a follow-up question about what governs that variation.

4.2.4 How the sensitivity varies with dataset structure. Do simple structural features of the practice-length distribution predict the magnitude or shape of the sensitivity? We examine this in two ways.

Sensitivity magnitude is positively associated with the heaviness of the practice-length distribution. Across the 23 datasets, L_{sd} correlates monotonically with the percent change in estimated student-slope IQR under $L \leq 10$ truncation (Spearman $\rho = 0.60$, $p = 0.002$; Figure 6). L_{sd} thus offers a coarse, model-free indicator of how susceptible a given dataset’s variance-component estimates are to the kind of sensitivity we document, available before any modeling decision.

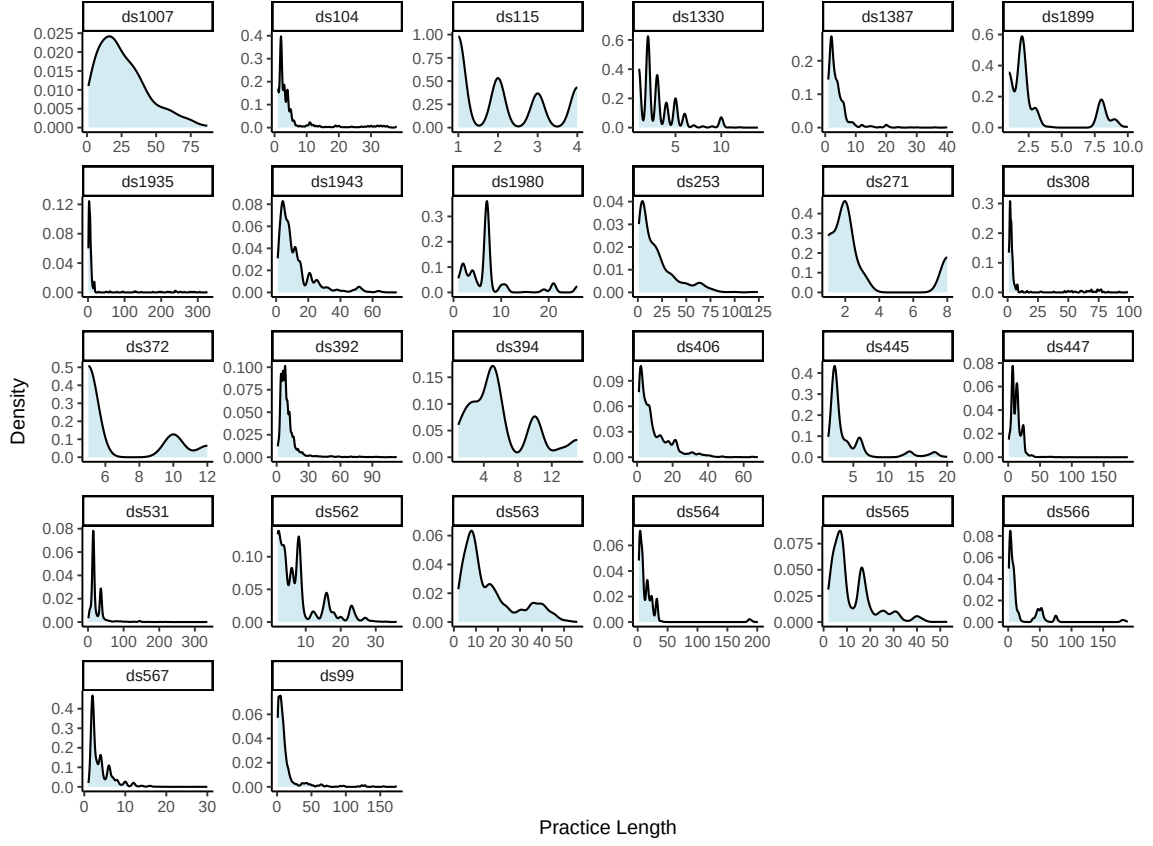


Figure 2: Distribution of student–KC pair lengths across all 26 datasets.

Sensitivity direction, however, is not uniformly predicted by structural features. Five datasets (ds1007, ds1330, ds406, ds562, ds567) show estimated IQR at $L \leq 10$ below the full-data estimate, the opposite of the modal pattern of inflation. The remaining 18 show inflation, but with non-monotonic curves of varying shape across k . The five exceptions are not a structurally coherent group: their median pair lengths range from 2 to 21, the proportion of pairs longer than 10 opportunities ranges from 0.2% to 78%, and dataset sizes range from 195 to over 45,000 student–KC pairs. No single structural feature reliably distinguishes the two groups at this sample size (Mann-Whitney U , all $p > 0.07$; largest effect size L_{sd} at $r = 0.54$, $p = 0.15$). The dataset-specific direction of the truncation effect thus appears to be governed by features of the data-generating process that are not summarized by the simple distributional statistics we examined.

Together, these results provide a partial picture: L_{sd} is informative about the magnitude of sensitivity, but no structural feature we examined predicts its direction or detailed shape. The consistent signal across the collection is the instability itself.

5 Discussion

Estimates of student learning-rate variability from the iAFM applied to the 26 datasets analyzed by Koedinger et al. [20] depend substantially on the structure of practice sequence lengths in the

underlying data. The empirical magnitude of this dependence diverges substantially from what standard estimation theory under ignorable truncation would predict, and its dataset-specific shape does not map onto simple structural features of the data. We discuss below the three candidate mechanisms introduced in Section 2.2—informative observation length, functional-form misspecification, and identification weakness at shallow depth—each of which could produce the kind of pattern we observe in some datasets. The heterogeneity across datasets is itself consistent with no single mechanism operating uniformly.

Informative observation length. In observational student log data, how much practice each student gets on a skill is shaped by mastery policies, adaptive routing, and student behavior. Mastery-based systems advance students once competence is demonstrated [6, 13]; students who struggle accumulate many attempts on a single skill without improving [5]; engagement varies with performance and difficulty [2, 3]. Each of these mechanisms ties practice length to the underlying learning trajectory—the missing-not-at-random condition under which standard mixed-effects estimation can yield biased variance components [10, 22, 23]. Methods for principled estimation under MNAR—joint longitudinal-time-to-event models, pattern-mixture models, selection models—would directly address the concern; we describe these in Section 5.2.

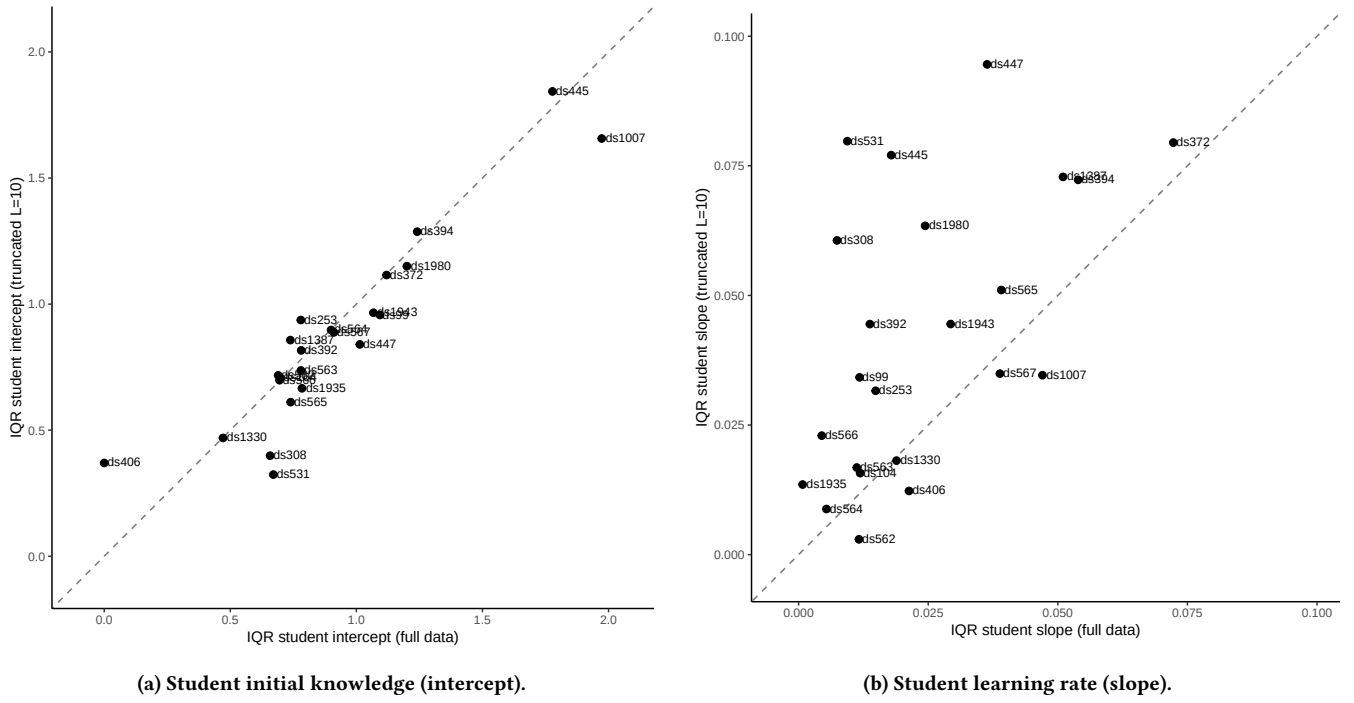


Figure 3: IQR of student random effects from the full model (x -axis) versus the truncated model ($L \leq 10$, y -axis). Each point represents one dataset; the diagonal indicates perfect agreement.

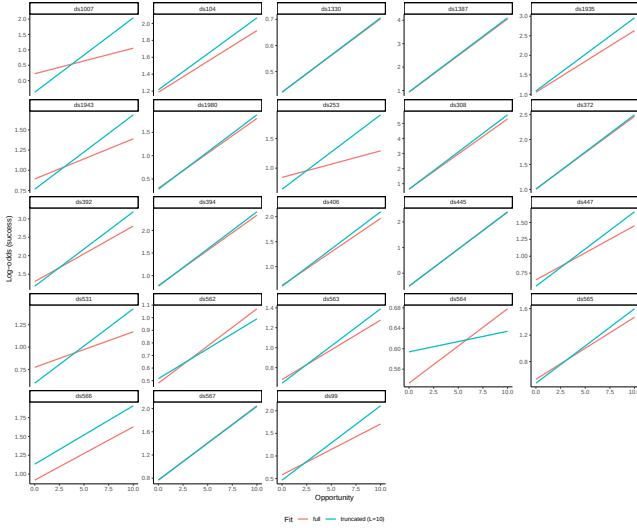


Figure 4: Estimated fixed effects (population-average learning trajectory) from models fit on full (red) and truncated ($L \leq 10$, blue) data. One panel per dataset.

Functional-form misspecification. The iAFM assumes log-odds of correct response are linear in practice opportunities, an assumption that may be violated for extended sequences where learning has plateaued [12, 14, 25]. If true learning curves are concave and the

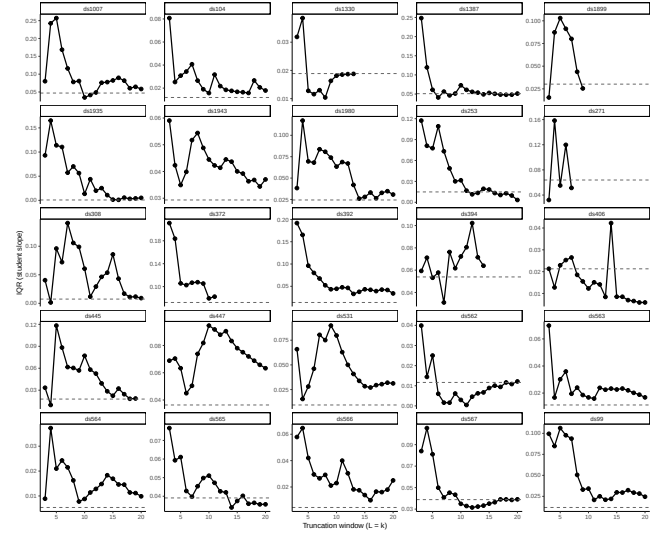


Figure 5: IQR of estimated student learning-rate slopes as a function of observation window k ($k = 5$ to $k = 20$) across 25 datasets (ds115 excluded; all its pairs have $L < 5$). Dashed horizontal lines indicate the IQR from the full, untruncated model.

linear specification holds only locally, the best-fitting linear slope for any segment depends on which portion of the curve the data

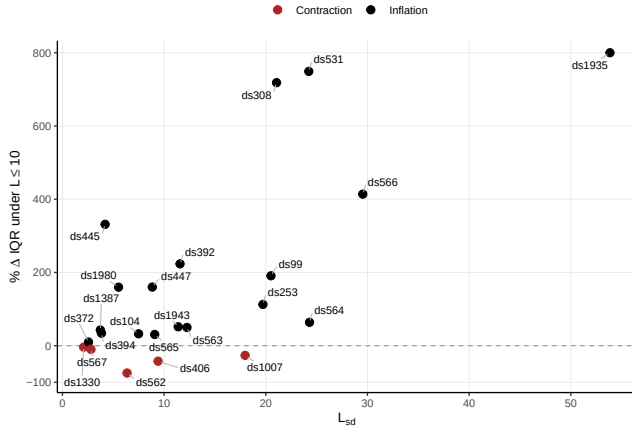


Figure 6: Practice-length distribution heaviness (L_{sd} , x-axis) versus the percent change in estimated student learning-rate IQR under $L \leq 10$ truncation (y-axis). One point per dataset; 23 datasets included. Five datasets (red) show contraction rather than inflation. The y-axis is capped at 800%; ds1935 is plotted at the cap, with a true value of +1661% reflecting a numerator-denominator artifact of an unusually small full-data IQR. Spearman $\rho = 0.60$ ($p = 0.002$), computed on raw uncapped values.

covers, which can produce systematic differences in estimated heterogeneity under truncation. Refitting with a nonlinear opportunity term and assessing whether the truncation sensitivity persists is a productive next step under this account.

Identification weakness at shallow depth. Liu and Koedinger [24], in introducing the iAFM, observed that dataset depth—the number of observations per student–KC pair—is critical for reliably estimating per-student parameters, and that sparse data degrade recovery. The median pair length across the 26 datasets is 6.5 opportunities. At such depths, the model has limited information from which to estimate per-student slopes separately from the population mean and the KC-level structure, regardless of whether the observation mechanism is ignorable or not. The resulting estimates are then dominated by the prior and the KC-level structure rather than by per-student information. Truncation amplifies this concern by further reducing per-pair information. This account predicts that sensitivity should be greater in datasets with heavier-tailed practice-length distributions, which the L_{sd} correlation in Section 4.2 is consistent with. It does not, however, account for the dataset-specific direction of the effect.

What we have shown empirically is the sensitivity itself: variance-component estimates in this modeling framework, on these data, depend substantially on observation depth. The three mechanisms above are the candidate accounts adjacent literatures provide; each is a plausible explanation for what we observe, and the dataset-specific heterogeneity in our findings is consistent with multiple mechanisms contributing across the collection rather than any single one. The robustness checks accompanying the original finding (simulation studies under amplified slope variability, replication

across datasets [31], bootstrapped confidence intervals) establish that the iAFM can detect heterogeneity when it is large; our analyses are consistent with that. They do not, however, address how variance-component estimates depend on the observation structure of the data when heterogeneity is at empirically realistic levels. That dependence is what we document.

The dependence these findings illustrate is not specific to the iAFM or to this dataset collection. It applies in principle to any mixed-effects student model estimated from observational learning data where practice exposure is not experimentally controlled. Its severity varies with the structure of the practice-length distribution: as Section 4.2 shows, L_{sd} is positively associated with the magnitude of sensitivity, though the direction and detailed shape of the effect are not summarized by simple distributional statistics. Reporting L_{med} and L_{sd} alongside model results provides a low-cost transparency standard that makes the relevant data structure visible to readers and reviewers.

Comparisons of learning-rate estimates across groups or conditions deserve particular caution. When practice exposure differs systematically across compared groups—for example, when a system routes struggling students to more practice—the length distributions for the groups will differ in ways correlated with the parameters being compared. Reported differences in estimated learning efficiency between instructional conditions or student populations may therefore reflect differences in practice exposure rather than differences in learning processes, and cannot be cleanly attributed to the intervention without accounting for how the practice sequences were generated. This concern extends beyond research claims to evaluations of tutoring systems and interventions that rely on variance components for personalization or fairness assessment: when exposure structure is not held constant, conclusions drawn from these estimates may reflect features of the routing system rather than features of the learners. Practice sequence length distributions should be reported alongside model fit, KC model quality, and sample size as part of standard methodological description.

5.1 Limitations

We note four considerations bearing on the scope of these analyses.

First, the present analyses document the sensitivity and identify three candidate mechanisms; they do not adjudicate among them, and they do not establish empirically that random data reduction can be ruled out as a contributor. Whether and in which direction the full-data estimate is biased relative to the population variance is a question observational analysis on these data alone cannot answer; the methods in Section 5.2 are the means by which it could be addressed.

Second, all fits use `lme4::glmer` with the same optimizer and an unstructured covariance specification. Mixed-effects fits with rich random-effect structures can be sensitive to optimizer choice and prone to singular-fit warnings. We did not systematically vary the optimizer or compare to alternative implementations (e.g., `glmmTMB`, Bayesian hierarchical fits with informative priors). Whether the truncation sensitivity we document is robust across estimation backends is a question we leave to follow-up work.

Third, the structural-features follow-up in Section 4.2 is exploratory and statistically underpowered. With $N = 23$, the Mann-Whitney tests can only detect large structural separations between inverted and non-inverted groups; we cannot rule out structural patterns that exist but require more datasets to detect. The reported negative finding is best read as “simple distributional statistics do not, on the present sample, account for the dataset-specific direction of the effect,” not as “no structural pattern exists.”

Fourth, the analyses draw on a single modeling framework, the iAFM, and one dataset collection. Whether the patterns generalize to other frameworks, including Bayesian knowledge tracing [8], deep knowledge tracing [26], and their extensions, remains an open empirical question.

5.2 Future Directions

The empirical findings here motivate three concrete directions for follow-up work.

Joint and pattern-mixture modeling. Joint models of longitudinal outcomes and time-to-event processes [28, 32, 36] simultaneously characterize the outcome trajectory and the observation process, allowing variance components to be estimated under explicit assumptions about why observation lengths vary. Adapting these methods to student log data—where the “event” is skill exit, whether through mastery, wheel-spinning, or disengagement—would yield variance-component estimates that account for non-ignorable observation length, and would allow the question of how much students differ in learning rate to be addressed under a clearer assumption set than maximum-likelihood estimation on the outcome alone. Pattern-mixture models [21], which stratify by observed sequence length and aggregate within-stratum estimates, offer a complementary approach under weaker structural assumptions.

Targeted simulation studies. The candidate mechanisms identified in Section 5 generate different mappings from latent learning rate to observed sequence length and to the iAFM’s estimates. Simulating data under each mechanism with known generative parameters, then refitting the iAFM under the truncation manipulation we use here, would establish whether the IQR inflation we report can be reproduced by each mechanism in isolation, and whether their relative magnitudes are diagnostic. Such simulations would also map the inflation factors observed here onto the magnitude of underlying bias in variance components.

Experimental control on practice exposure. The cleanest path to ground truth is to break the dependence between learning rate and sequence length by design. Studies in which students complete a fixed, predetermined number of opportunities per skill— independent of mastery state, performance, or engagement—would yield estimates of learning-rate variability that are not subject to informative observation length. Even partial controls, such as a randomized minimum number of opportunities, would provide useful benchmarks against which observational estimates from the same population could be calibrated.

These directions are complementary rather than alternative: simulation characterizes bias under specific mechanisms, joint and pattern-mixture models provide observational corrections, and experimental data anchor both.

What the present analyses establish is more modest. The opening question—how much do students vary in their rate of learning?—turns out to depend on a methodological choice the original analysis did not need to make explicit: how deep into each student’s practice sequence the model is allowed to look. We have shown that conclusions about student learning-rate heterogeneity drawn from observational data depend on the observation structure of those data in ways that have gone unexamined, and that the magnitude of this dependence is large. Whether the original variance-component estimates reflect students’ actual learning-rate variability or a methodological signature of how the data were generated is the natural follow-up question, and one that observational analysis on these data alone cannot resolve. Answering it requires treating practice sequence length distributions not as incidental features of data collection but as a substantive property of the estimation problem, one that must be examined, reported, and accounted for before conclusions about population-level learning dynamics can be drawn with confidence.

References

- [1] Vincent Alevan, Elmar Stahl, Silke Schworm, Frank Fischer, and Raven Wallace. 2003. Help seeking and help design in interactive learning environments. *Review of Educational Research* 73, 3 (2003), 277–320.
- [2] Ryan S. J. d. Baker. 2007. Modeling and understanding students’ off-task behavior in intelligent tutoring systems. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems*. 1059–1068.
- [3] Ryan S. J. d. Baker, Albert T. Corbett, Kenneth R. Koedinger, Shelley Evenson, Ido Roll, Angela Z. Wagner, Meghan Naim, Jay Raspat, Daniel J. Baker, and Joseph E. Beck. 2006. Adapting to when students game an intelligent tutoring system. In *Proceedings of the International Conference on Intelligent Tutoring Systems*. Springer, 392–401.
- [4] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67, 1 (2015), 1–48.
- [5] Joseph E. Beck and Yue Gong. 2013. Wheel-spinning: Students who fail to master a skill. In *Proceedings of the International Conference on Artificial Intelligence in Education*. Springer, 431–440.
- [6] James H. Block and Robert B. Burns. 1976. Mastery learning. *Review of Research in Education* 4 (1976), 3–49.
- [7] Hao Cen, Kenneth Koedinger, and Brian Junker. 2006. Learning Factors Analysis: a general method for cognitive model evaluation and improvement. In *Proceedings of the International Conference on Intelligent Tutoring Systems*. Springer, 164–175.
- [8] Albert T. Corbett and John R. Anderson. 1994. Knowledge tracing: modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction* 4, 4 (1994), 253–278.
- [9] Michael J. Daniels and Joseph W. Hogan. 2008. *Missing Data in Longitudinal Studies: Strategies for Bayesian Modeling and Sensitivity Analysis*. Chapman and Hall/CRC.
- [10] Peter Diggle and Michael G. Kenward. 1994. Informative drop-out in longitudinal data analysis. *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 43, 1 (1994), 49–73.
- [11] Shayan Doroudi and Emma Brunskill. 2017. The misidentified identifiability problem of Bayesian knowledge tracing. In *Proceedings of the 10th International Conference on Educational Data Mining (EDM)*.
- [12] Nathan J. Evans, Scott D. Brown, Douglas J. K. Mewhort, and Andrew Heathcote. 2018. Refining the law of practice. *Psychological Review* 125, 4 (2018), 592–605.
- [13] Theodore W. Frick. 1990. A comparison of three decision models for adapting the length of computer-based mastery tests. *Journal of Educational Computing Research* 6, 4 (1990), 479–513.
- [14] Andrew Heathcote, Scott Brown, and D. J. K. Mewhort. 2000. The power law revealed: the case for an exponential law of practice. *Psychonomic Bulletin & Review* 7, 2 (2000), 185–207.
- [15] Joseph W. Hogan, Jason Roy, and Christina Korkontzelou. 2004. Handling dropout in longitudinal studies. *Statistics in Medicine* 23, 9 (2004), 1455–1497.
- [16] Qi Huang, Daniel M. Bolt, and Xiangyi Liao. 2025. Theory-driven IRT modeling of vocabulary development: Matthew effects and the case for unipolar IRT. *Journal of Educational Measurement* (2025).
- [17] Tanja Käser, Kenneth R. Koedinger, and Markus Gross. 2014. Different parameters – same prediction: an analysis of learning curves. In *Proceedings of the 7th International Conference on Educational Data Mining*. 52–59.

- [18] Kenneth R. Koedinger, Ryan S. J. d. Baker, Kyle Cunningham, Alida Skogsholm, Brett Leber, and John Stamper. 2010. A data repository for the EDM community: the PSLC DataShop. In *Handbook of Educational Data Mining*, Cristóbal Romero, Sebastián Ventura, Mykola Pechenizkiy, and Ryan S. J. d. Baker (Eds.). CRC Press, 43–56.
- [19] Kenneth R Koedinger, Emma Brunskill, Ryan Sjd Baker, Elizabeth A McLaughlin, and John Stamper. 2013. New potentials for data-driven intelligent tutoring system development and optimization. *Ai Magazine* 34, 3 (2013), 27–41.
- [20] Kenneth R. Koedinger, Paulo F. Carvalho, Ran Liu, and Elizabeth A. McLaughlin. 2023. An astonishing regularity in student learning rate. *Proceedings of the National Academy of Sciences* 120, 13 (2023), e2221311120. doi:10.1073/pnas.2221311120
- [21] Roderick J. A. Little. 1993. Pattern-mixture models for multivariate incomplete data. *J. Amer. Statist. Assoc.* 88, 421 (1993), 125–134.
- [22] Roderick J. A. Little. 1995. Modeling the drop-out mechanism in repeated-measures studies. *J. Amer. Statist. Assoc.* 90, 431 (1995), 1112–1121.
- [23] Roderick J. A. Little and Donald B. Rubin. 2019. *Statistical Analysis with Missing Data* (3rd ed.). John Wiley & Sons.
- [24] Ran Liu and Kenneth R. Koedinger. 2017. Towards reliable and valid measurement of individualized student parameters. In *Proceedings of the 10th International Conference on Educational Data Mining*. 135–142.
- [25] Allen Newell and Paul S. Rosenbloom. 1981. Mechanisms of skill acquisition and the law of practice. In *Cognitive Skills and Their Acquisition*, John R. Anderson (Ed.). Psychology Press, 1–55.
- [26] Chris Piech, Jonathan Bassen, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas J. Guibas, and Jascha Sohl-Dickstein. 2015. Deep knowledge tracing. In *Advances in Neural Information Processing Systems*, Vol. 28.
- [27] Stephen W. Raudenbush and Anthony S. Bryk. 2002. *Hierarchical Linear Models: Applications and Data Analysis Methods* (2nd ed.). Sage Publications.
- [28] Dimitris Rizopoulos. 2012. *Joint Models for Longitudinal and Time-to-Event Data: With Applications in R*. CRC Press.
- [29] G. K. Robinson. 1991. That BLUP is a good thing: the estimation of random effects. *Statist. Sci.* 6, 1 (1991), 15–32.
- [30] Donald B. Rubin. 1976. Inference and missing data. *Biometrika* 63, 3 (1976), 581–592.
- [31] Mary Ann Simpson, Kole A. Norberg, and Stephen E. Fancsali. 2024. Replicating an "Astonishing Regularity in Student Learning Rate". In *Proceedings of the 17th International Conference on Educational Data Mining*. 420–425.
- [32] Anastasios A. Tsiatis and Marie Davidian. 2004. Joint modeling of longitudinal and time-to-event data: an overview. *Statistica Sinica* 14 (2004), 809–834.
- [33] Kurt VanLehn. 2006. The behavior of tutoring systems. *International Journal of Artificial Intelligence in Education* 16, 3 (2006), 227–265.
- [34] Jaroslav Čechák and Radek Pelánek. 2019. Item ordering biases in educational data. In *Proceedings of the International Conference on Artificial Intelligence in Education (Lecture Notes in Computer Science)*. 48–58.
- [35] Herbert J Walberg and Shiow-Ling Tsai. 1983. Matthew effects in education. *American educational research journal* 20, 3 (1983), 359–373.
- [36] Lang Wu, Wei Liu, Grace Y. Yi, and Yangxin Huang. 2012. Analysis of longitudinal and survival data: joint modeling, inference methods, and issues. *Journal of Probability and Statistics* 2012 (2012), 640153.
- [37] Margaret C. Wu and Raymond J. Carroll. 1988. Estimation and comparison of changes in the presence of informative right censoring by modeling the censoring process. *Biometrics* 44, 1 (1988), 175–188.